



Deep multi-view document clustering with enhanced semantic embedding

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ABSTRACT

Multi-view clustering, which aims to group data with multiple views, has recently attracted intense research attention. Text documents bring additional difficulties to multi-view clustering due to the sparseness, high dimensionality, and inconsistency of document views. In this paper, we introduced a novel model on multi-view document clustering with enhanced semantic embedding, namely, MDCE, to address all of the above difficulties of clustering text documents with more than one representation view. Enhanced semantic embedders are designed to learn and improve the semantic mapping from higher-dimensional document space to lower-dimensional feature space with complementary semantic information. Specifically, three types of complementary semantic information are involved in an unsupervised manner: neighbour-wise, view-wise, and cluster-wise complementary information. A deep network is designed to optimize the enhanced semantic mapping, integrate lower-dimensional features from multiple views, and discover document clustering assignments simultaneously. We conducted extensive experiments on our proposed MDCE model by using realistic datasets compared with a number of state-of-the-art multi-view clustering approaches. Experimental results demonstrate that the MDCE-related models perform substantially better than all other models.

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1. Introduction

Unsupervised document clustering is an important task in text mining and machine learning. Traditional document clustering deals with documents with a single set of features. With the rapid development of internet technology, many real-world documents appear in multiple views. For example, news articles can be expressed not only by the features from the traditional content view but also by a set of propagation behavior features, such as the readers and the forwarders of the news articles. A research paper can be described by the set of discriminative content features or by the views of researchers, including paper authors, reference authors, and citation authors. These multiple views of text documents create new opportunities and challenges in document clustering.

Multi-view clustering, which aims to integrate multiple views to help identify data grouping structures, is receiving increasing research interest. There has been little research work on multi-view clustering of text documents. Compared with

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other real-world data, text documents with multiple views present a number of unique difficulties. 1) Text documents have sparse representation, as the number of discriminative features in a text document is limited, and crucial discriminative features may be missing. The sparseness problem in document representation leads to inaccurate estimation of document assignments. Similar documents are frequently regarded as not close and organized in different data groups. It is useful to discover these missing key semantic features and complement the document view automatically. The document clustering problem is unsupervised. The complementary process needs to be conducted without requiring that key features of the document be labelled. 2) Each view of text documents is high-dimensional. The curse of dimensionality can make it difficult to separate documents in high-dimensional space. The curse of dimensionality leads to poor clustering performance because the clustering process relies heavily on measuring the distance of data samples to determine the relative relationships of data samples. The curse of dimensionality is aggravated when multiple high-dimensional views are integrated. Therefore, it is crucial to map the high-dimensional feature space of the original document space to the lower-dimensional feature space. 3) Multi-views of text documents are inconsistent. Documents are organized differently in different views. Multi-view document clustering needs to balance multiple document views and organize documents in a unified structure.

We design a multi-view document clustering model to deal with all of the above issues. The higher-dimensional document feature space is mapped and fused to lower-dimensional feature spaces enhanced by additional semantic information in an unsupervised manner. There are three types of complementary semantic information: neighbour-wise complementary information, view-wise complementary information, and cluster-wise complementary information. First, close neighbours of documents commonly carry complementary semantic information. Documents share key features when they are close. Close neighbours of a document normally contain other key features missed by the current documents. Second, complementary semantic information is available across multiple views of documents. Multiple views of documents have complementary effects on the discovery of document structure. Document clustering frequently may perform poorly in one view but give good clustering results on other views. For example, news that is of interest to similar people generally has similar themes. However, features in the content view of these news articles are often different due to differences in author writing style or the evolution of news events. It would be useful to capture the cross-view complementary semantic information to enhance learning of the mapping of semantic lower-dimensional feature space for document views. Finally, the document clustering process also has complementary semantic information. In the clustering process, documents are iteratively assigned to clusters with close centroids. A cluster collects information from all its members. Ideally, a cluster can be regarded as containing nearly all key semantic features of the underlying cluster topic, while a document of a cluster is a sample of the underlying topic that contains part of the key features. Therefore, it is useful to discover all three types of semantic information and improve the mapping of document space to lower-dimensional feature space to solve the sparseness and high-dimensionality problem of multi-view document clustering. Moreover, capturing cross-view semantics also helps unify the features from different views, thus solving the inconsistency problem.

In this paper, we propose a novel deep multi-view document clustering model with enhanced semantic embedding, namely, the MDCE model, to cluster documents with multiple views. The main contributions are summarized as follows:

- Enhanced semantic feature embedders are designed to automatically discover and complement feature mapping with neighbour-wise and view-wise complementary information in an unsupervised manner.
- View-specific enhanced semantic mapping and multi-view representation learning are jointly performed, and the clustering process is used for fine tuning.
- A three-layer-deep model containing ESEs, a fusion layer and a clustering layer called MDCE is developed to deal with the sparse, high-dimensional, and inconsistency problems of multi-view document clustering.

We conducted extensive experiments using realistic datasets to compare our proposed MDCE model with a number of state-of-the-art document clustering approaches. The experimental results demonstrate that the MDCE-related models are effective and substantially improves document clustering performance.

The remainder of this paper is organized as follows. Section 2 reviews related work on multi-view clustering. In Section 3, we describe our model in detail. Section 4 presents the experimental results. We present conclusions and future work in Section 5.

2. Related work

This paper is closely related to three branches of study, multi-view representation, multi-view clustering and deep joint clustering. In the following paragraphs, we give series of work that lead to our own.

Multi-view representation Multi-view representation learning addresses the problem of learning representations (or features) of the multi-view data to facilitate the extraction of useful information for developing prediction models [1]. One typical direction of multi-view representation models is to learn latent factors for better capturing data's underlying representative characteristics. The latent factors can be regarded as reflecting the structure of dataset with multiple views. We summarized the models proposed to find the data's structure by learning a better multi-view data representation. In general, there are three main ways on these multi-view representation models, in particular, probabilistic based models, subspace based models, and deep neural based models. Probabilistic based on multi-view representation models attempt to

construct generative probabilistic models to learn the underlying data structures. [2] proposed an extended LDA model with multi-view data, namely Corr-LDA. The Corr-LDA model simultaneously conducts dimensionality reduction and models the conditional correspondence between the reduced and original representations. MVPLSA [3] is a generative model for multi-view learning. It uses the probabilistic latent semantic analysis to jointly modeling the co-occurrences of features and documents from different views. Models on subspace-based representation learning are used to discover more clearly data structures on lower dimensional subspaces. Multi-co-training(MCT) [4] is a semi-supervised learning method that attempts to exploit various perspectives in terms of view subsets for the same example. With the introduction of a kernel dependence measure, FMR [5] enforces a single view as close to other views to flexibly encode complementary information from different views and explores nonlinear correlations among these views. The FMR model is related to the model proposed in this paper as it also makes use of information across data views to complement the view representations. In addition to the probabilistic-based and subspace-based models, there are a number of deep neural-based multi-view representation methods. DBM [6] was presented to deal with multiple modalities. It combines image-specific and text-specific layers with an additional layer of binary hidden units. [7] was proposed to extract shared representations by training a bimodal deep autoencoder. It exploits the concatenated final hidden codings of audio and video modalities as input and maps these inputs to a shared representation layer. [8,9] are derived based on [7]. In addition to cross-mapping, they add a reconstruction of current view to preserve local structures. The main motivation of all the above multi-view representation models is to learn consistence representations of the data samples. The underlying data structures captured by the multi-view data representations can be regarded as secondary results. However, there is little evidence that the data partition indicating by the multi-view data representation is the same with the optimal solution of data assignments. Moreover, the sparse, the high-dimensional, and the inconsistent problems of multi-view document clustering are not addressed directly. In this paper, we attempt to address all above problem for text documents with the motivation of discovering the optimal document partition solutions.

Multi-view clustering Clustering data with multiple views is of great research interest. Taking the consistency on the solution of cluster assignment into consideration, clustering multi-view data can be further divided into several tasks. The first task, namely multi-view multiple clustering (MVMC) [10,11], is derived from the task of multiple clustering (MC) which aims at exploring alternative ways to organize data samples into meaningful groups from different perspectives. Models of MVMC discover interesting but related clustering results for each data view. It allows inconsistent cluster assignments. Another related but different task is multiple co-clustering task (MultiCC) which is developed based on both the MC and the co-clustering problem. Other than generating one-way clustering result, the MultiCC task captures multiple two-way clustering results including not only the sample cluster structure but also the feature cluster structure simultaneously [12,13]. On the contrary of the multi-view clustering problem, the MultiCC task targets at single-view data. In this paper, we are more focus on the task of multi-view clustering (MVC), which aims to classify similar data samples into the same group by combining multi-view feature information and discover consistent clustering partition across different views [14]. In the rest part of this paragraph, we illustrated related research works on MVC in detail. [15] first proposed clustering with multi-view data in 2004. This research attempted to solve the multi-view clustering problem using an EM algorithm. Based on the convex mixture models for single-view clustering, the multi-view version proposed in [16] avoids the initialization and local optima problems of standard mixture models by locating the global optimum. These methods are called generative MVC approaches. There is another type of methods that directly optimize the model objective to seek the best clustering solution, which is called a discriminative method. Developing discriminative models for MVC has attracted much attention in recent years. The main idea of these methods is to learn a common low dimensional space for multiple views with the purpose of achieving consistent clustering results. Various research efforts have been made. Along this line, [17] [18] used the canonical correlation analysis (CCA) to project the multi-view high-dimensional data onto a low-dimensional subspace for downstream clustering. [19] [20] [21] employed nonnegative matrix factorization (NMF) to seek common latent factors among multiple views. A deep structure was proposed to conduct Semi-NMF hierarchically to learn the hierarchical semantics of multi-view data in a layer-wise fashion [19]. [20] also designed a hierarchical MF-based approach for the MVC problem. A collaborative deep matrix decomposition framework is employed to uncover the hierarchical semantics of the input data in a layer-wise way. MVOCNMF [21] is a semi-supervised multi-view clustering approach that learns the low-dimensional representations of data by the constrained-NMF technique based on the label information. There are also some MVC research works investigating on the subspace clustering models [22,23]. In [22], a series of multi-view subspace clustering model (MLRSSC) are investigated to learn a joint subspace representation by constructing affinity matrix shared among all views. The objective of the MLRSSC models are developed to balance between the agreement across different views, while at the same time encourages sparsity and low-rankness of the solution. [23] introduces a multi-view subspace clustering model to construct an intactness-aware similarity based on the intact space learning technique. It learns an intact space by integrating encoded complementary information.

Deep joint clustering Recently, an increasing number of studies have suggested jointly learning feature representation and clustering via neural networks. A deep clustering network (DCN) [24] is a joint dimensional reduction and k-means clustering framework where the dimensional reduction model is investigated based on a deep neural network. [25] presented a deep embedded clustering (DEC) method that learns a mapping from the data space to a lower-dimensional embedded space in which it iteratively optimizes a KL divergence-based clustering objective. The IDEC [26] model was designed by adding

local structures on the basis of DEC. Deep joint clustering has also been investigated for the problem of clustering multi-view data. A normal strategy for it is to learn a unified data representation shared by all multiple views via deep neural networks for conducting the clustering process. According the clustering model employed, existing deep joint clustering models for multi-view data can be further categorized into three different categories, in particular, deep multi-view clustering based on k-means, deep multi-view clustering based on DEC, and multi-view subspace clustering. In the first category, deep neural networks are linked with the traditional k-means model for discovering data structure [27] [28]. Alternating-view k-means (AV-kmeans) [27] model was designed to investigate the problem of user question/query intent induction. It computes the cluster assignment on one view with a k-means style mechanism and then train the encoder of the other view. The AE²-Nets [28] model jointly learns the representation of each view and encodes them into an intact latent representation with a nested autoencoder framework. The learned latent representation is used for various of subsequent tasks which are incorporated separately. k-means model is applied for the clustering problem in their investigation. There are an increasing number of multi-view clustering models designed based on the DEC model [25] because it shows promising results on optimizing clustering assignment and data embedded representation simultaneously. Therefore, we organize these models into the second category. [29] extends the DEC model to multi-view data. A joint framework of deep multi-view clustering (DMJC) is proposed which learns the multiple deep embedded features, multi-view fusion mechanism and clustering assignment simultaneously. DAMC [30] adopts deep auto-encoders to learn latent representations shared by multiple views, and meanwhile leverages adversarial training to further capture the data distribution and disentangle the latent space. Other than directly use the KL divergence loss function employed by the DEC model, it designs a loss function that incorporate the autoencoder loss and the generative adversarial network loss as well. DEMVC [31] learns feature representation and clustering assignment of all views through collaborative training. The loss function of the DEMVC model is designed as a multi-view generalized version of IDEC. In [32], a task-guided One2Multi graph autoencoder clustering framework (O2MAC) is designed for the multi-view graph clustering problem. It utilizes one view graph convolutional encoder and multi-view graph structure decoder to encode the attributed multi-view graphs to a low-dimensional space. Inspired by DEC, the O2MAC designs a self-training clustering objective which jointly optimize the graph autoencoder embedding and clustering learning. In the third category, multi-view subspace clustering models are investigated. DMVSSC [33] extends the self-expressiveness based sparse subspace clustering to multi-view setting. It extracts multi-view deep features by CCA-guided convolutional auto-encoders and then learns a joint deep subspace self-expressive representation across all views. MvDSCN [34] learns a multi-view self-representation matrix in an end-to-end manner by combining convolutional auto-encoder and self-representation together. It employs a diversity net (Dnet) and a universality net (Unet) to learn view-specific self-representation matrices and common self-representation matrix. The S2DMVSC [35] model integrates spectral clustering and affinity learning into a deep learning framework.

All existing research works for multi-view clustering focus on learning the fusion of multi-view representations and discover consistent cluster assignments. There has been little work investigating the learning of the document view representations enhanced by additional semantic information. [23] uses the neighbors for each intact space data point to adaptively construct the data similarity matrix. APMC [36] uses anchors to reconstruct instance-to-instance relationships to integrate intra- and inter-view similarities. FMR [5] model the latent view representation by encoding complementary information from different views and exploring various correlations among views. However, the above methods complement the data representation from the perspective of getting a consistent representation across data views. None of them consider the data sparseness problem of text documents which leads to the lack of discriminative features for getting good clustering assignments. Furthermore, none of them explored the complementary information systematically. In this paper, we design a multi-view document clustering model taking the problems of data sparseness, high-dimensionality and inconsistency into consideration. We explore different types of complementary semantic information to improve the mapping of document space to lower-dimensional feature space. A deep network is designed to simultaneously optimize the enhanced semantic mapping, multi-view representation fusion, and the document clustering.

3. The MDCE model

In this section, we describe our MDCE model in detail. The MDCE model is composed of two parts. The first part is the enhanced semantic embedders (ESEs), which learn enhanced semantic mapping from higher-dimensional document spaces to lower-dimensional feature space. The ESEs handles two problems of multi-view document clustering, in particular, the high-dimensionality problem and the sparseness problem. The high-dimensionality problem is addressed by discovering the lower-dimensional feature space. The sparseness problem of text documents is solved by involving additional semantic information to enhance the representation of each document view. Different strategies are employed considering various complementary semantic information. In particular, the neighbour-based autoencoder model (NAE) and the cross-view autoencoder model (CAE) are designed to involve neighbour-wise and view-wise complementary information, respectively.

After the ESEs discover the appropriate mapping for each document view, they are then used as input in the second part of the MDCE model, which is a multi-channel deep document clustering model (MDCD). The MDCD model learns the document assignment, fuses the multiple views of document features, and simultaneously refines the feature mapping with cluster-wise complementary information. A unique optimal clustering assignment will be discovered by the MDCD model which solves the inconsistency problem of multi-view document clustering.

3.1. Enhanced semantic embedders

In this section, we introduce the design of two types of ESEs. We adopt feature embedding autoencoders for designing various types of ESEs to map the higher-dimensional document feature space to the lower-dimensional feature space. Two autoencoder mapping strategies are designed. The first strategy is the NAE model, which is designed to learn a single view of a document involving neighbour-wise complementary semantic information. The second strategy is the CAE model, which is designed to learn the low-dimensional feature space considering the view-wise complementary semantic information.

3.1.1. Neighbour-based autoencoder model

The NAE model uses the neighbours of each document to adjust the learning of the feature space. Given a representation of a document view, x^v , the NAE model captures the K nearest neighbours of x^v . The deep autoencoder network structure of the NAE model is illustrated in Fig. 1.

The NAE model is designed based on the deep AE network structure, because it shows promising results on learning semantically meaningful representations for real data and also can preserve local structure of data generating distribution [37,42]. The NAE model is trained layer by layer. Each layer is used to reconstruct the output of its previous layer. Because the representation of documents is sparse, we keep all features of a document view by setting each layer of NAE with an under-complete autoencoder as follows:

$$\begin{aligned} z_n^v &= g_e^n(W_e^n x^v + b_e^n) \\ \hat{x}^v &= g_d^n(W_d^n z_n^v + b_d^n) \end{aligned} \quad (1)$$

where g_e^n and g_d^n are activation functions for the encoding and decoding network, respectively, of the NAE network. x^v is the input of the autoencoder, indicating the original representation of document x with view v . $\hat{x}^v$ is the output of the autoencoder, indicating the construction representation of x^v . The training is conducted by minimizing the loss function, defined as follows:

$$L^n(x^v; \Theta^n) = \|\eta_x^v - \hat{x}^v\|_2^2 \quad (2)$$

where $\|\cdot\|_2$ is the L_2 norm. $\Theta^n = \{W_e^n, b_e^n, W_d^n, b_d^n\}$ are NAE model parameters. η_x^v is the representation of the neighbours of document x^v . In the NAE model, η_x^v is learned by first selecting K neighbours of x^v with the same idea as the KNN model and then taking the average of those selected neighbours.

The NAE model is given by Algorithm 1.

Algorithm 1: NAE.

Input: Dataset $D = \{(x_1^v, x_2^v, \dots, x_N^v)\}_{v=1}^V$, where N is the number of samples, V denotes the different document views; the NAE model;

Output: Parameter-determined NAE for given document views v .

- 1 Select the number of close neighbours K in $[1, 60]$, given the *dis* function and D^v corresponding to given view v ;
- 2 **foreach** x^v in $D^v = \{x_1^v, x_2^v, \dots, x_N^v\}$ **do**
- 3 find the K samples nearest to x^v in view v of D according to *dis*, denoted as $\eta_K(x^v)$;
- 4 $\eta_x^v = \eta_K(x^v).mean()$;
- 5 **end**
- 6 Here, get the set $D'^v = \{(x_1^v, \eta_{x_1}^v), (x_2^v, \eta_{x_2}^v), \dots, (x_N^v, \eta_{x_N}^v)\}$;
- 7 Init NAE model parameters Θ^n ;
- 8 **while** not meet termination condition **do**
- 9 **foreach** $(x^v, \eta_x^v) \in D'^v$ **do**
- 10 compute $\hat{x}^v$ by formula (1);
- 11 compute $\nabla_{\Theta^n} L^n(\Theta^n; x^v, \eta_x^v)$;
- 12 update Θ^n ;
- 13 **end**
- 14 **end**

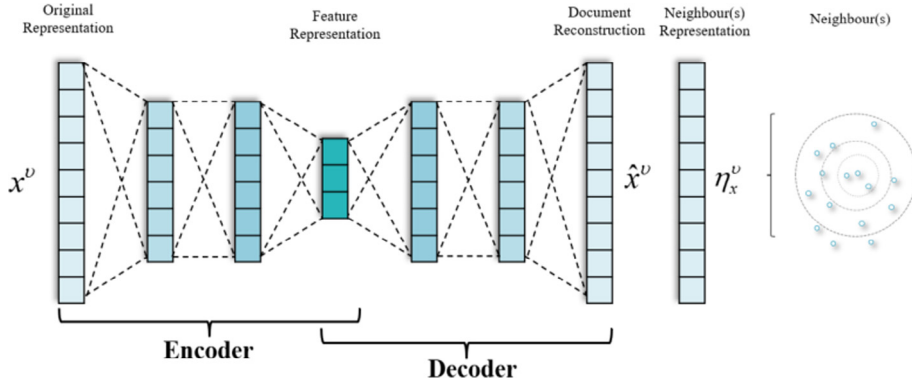


Fig. 1. The NAE network structure. A deep autoencoder is used to map the document x^v to a low-dimensional feature space in which x^v is semantically close to its selected neighbours. For each document, V NAEs are employed to capture its neighbour-wise complementary information.

After all the autoencoder layers are initially trained, the NAE model concatenates all the encoder parts of each autoencoder to form a deep encoder network. A deep decoder network is similarly constructed by concatenating all decoder parts of each autoencoder in reverse order. We use the deep encoder network of the NAE models in the following multi-channel deep document clustering model (MDCD) to map each view of the document in the document space to the lower-dimensional feature space.

3.1.2. Cross-view autoencoder model

The cross-view autoencoder model (CAE) model is designed based on the NAE model. In addition to the complementary information collected from the same-view neighbours, the CAE model collects complementary information cross-wise from other view representations. We assume that two documents x_i and x_j are related when they are close in most but not necessarily all document views. Inconsistent measurements of x_i and x_j indicate that the features in these views are incomplete and defective. Given a representation of a document x_i in the view v , denoted by x_i^v , a neighbour factor $\lambda_{x_i \rightarrow x_j}^v$ is designed to measure the inconsistency degree of document x_j to x_i in the document view space v :

$$\lambda_{x_i \rightarrow x_j}^v = \frac{s(x_i, x_j)^v}{\sqrt{\prod_{\tilde{v} \in -v} s(x_i, x_j)^{\tilde{v}}}} \quad (3)$$

where $s(x_i, x_j)^v$ is a traditional similarity measure calculating the degree of similarity between documents x_i and x_j under document view representation v . $-v$ denotes the set of view representations $\tilde{v}$ that $\tilde{v} \neq v$. Note that $\lambda_{x_i \rightarrow x_j}^v = \lambda_{x_j \rightarrow x_i}^v$.

In this paper, we use the cosine similarity to measure $s(x_i, x_j)^v$. $\lambda_{x_i \rightarrow x_j}^v$ detects documents that have inconsistent measurements across different views compared to document x_i . The lower the value of $\lambda_{x_i \rightarrow x_j}^v$, the more likely that x_j will be selected as a neighbour of x_i . By adopting $\lambda_{x_i \rightarrow x_j}^v$ in the KNN idea, K neighbours are selected for each document in each view and are used to learn the mapping of features through a deep autoencoder. The deep autoencoder is designed based on the network structure of the NAE model and is illustrated in Fig. 2.

The CAE model is trained layer by layer. Each layer is used to reconstruct the output of its previous layer. We trained the CAE model for every single-view representation and trained the parameter of the CAE by using the same set of neighbours for all other views. Each layer of the CAE is as follows:

$$\begin{aligned} z_c^v &= g_e^c(W_e^c x^v + b_e^c) \\ \hat{x}^{\tilde{v}} &= g_d^c(W_d^{\tilde{v}} z_c^v + b_d^{\tilde{v}}) \end{aligned} \quad (4)$$

where g_e^c and g_d^c are activation functions for the encoding and decoding network, respectively, of the CAE network. x^v is the input of the autoencoder indicating the original representation of document x with view v . $\hat{x}^{\tilde{v}}$ is the output of the autoencoder indicating the construction representation of x in all other views $\tilde{v}$ ($\tilde{v} \neq v$). Note that the encoder part of the autoencoder shares the same parameters, while the CAE uses different parameters to decode the intermediate result to other views. The training is conducted by minimizing the loss function, defined as follows:

$$L^c(x^v; \Theta^c) = \sum_{\tilde{v} \in -v} (\|\eta_x^{\tilde{v}} - \hat{x}^{\tilde{v}}\|_2^2) \quad (5)$$

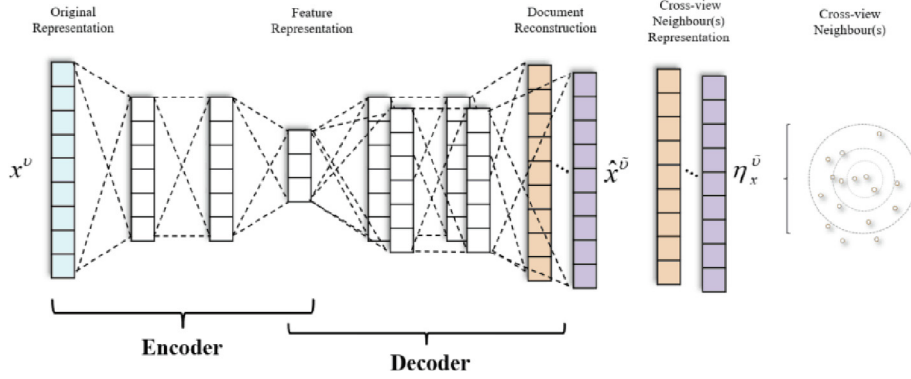


Fig. 2. The CAE network structure. A one-to-many deep autoencoder is used to map the document view x^v to a low-dimensional feature space in which all other views representations $\hat{x}^v$ are reconstructed. For each document, V CAEs are employed to capture its view-wise complementary information.

where $\|\cdot\|_2$ is the L_2 norm. Θ^c is the set of the CAE model parameters. η_x^v is the representation of the cross neighbours of document x^v .

The CAE introduces cross-view information, so in the algorithm, it is necessary to traverse for views other than v to obtain the detection factor $\lambda_{x_i \rightarrow x_j}^v$ and select a common K nearest neighbours as the representation of the other views, formed as (x^v, η_x^v) . It should be noted that the neighbor samples selected in each view using the similarity formula (3) may be different, so we use the trick of obtaining sample intersections to force each view to decode the same neighbor samples to achieve better capture of cluster information. The remaining steps are the same as NAE. More details can be found in Algorithm 1.

Similar to the NAE model, the CAE model fuses all encoder layers of each autoencoder to form an encoder network, and $V - 1$ decoder networks after the autoencoder layers are initially trained. We use the deep encoder network of the CAE models in the following MCDC model to map each view of the document in the document space to the low-dimensional feature space.

3.2. Multi-channel deep clustering model

By using the ESEs, each view of the document is mapped from the higher-dimensional document space to the lower-dimensional feature space. A multi-channel deep clustering model (MCDC) is then employed to simultaneously learn the document assignment and fine-tune the feature mapping with cluster-wise complementary semantics. The network structure of the MCDC model is illustrated in Fig. 3.

The input layer of the MCDC model is the original representation of each document view. In the embedding layer, ESEs, which are introduced in Section 3.1, are employed to map each document view to lower-dimensional feature spaces complemented by neighbour-wise or view-wise semantic information. Each encoder part of feature embedder is linked to the MCDC model as a single channel. Note that it is possible that one document view is embedded by more than one ESE with different types. The embedded multi-view document representations are then fused automatically in the fusion layer. There can be arbitrary design of the fusion layer. In this paper, we use a simple fusion strategy that does not introduce any

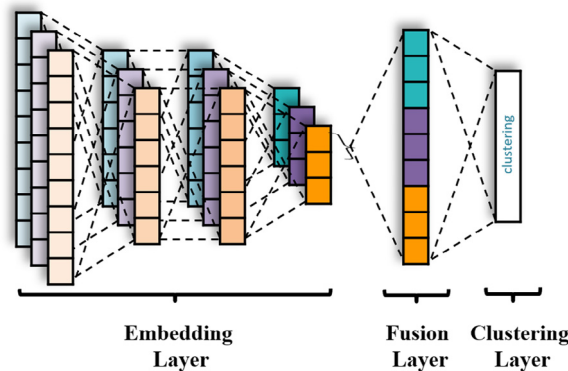


Fig. 3. The network structure of the MCDC model. The embedding layer contains a set of ESEs that are used to map the original document view to its latent representation. The fusion layer and the clustering layer are used to fuse multi-view representations and learn the cluster assignment, respectively.

additional parameters. It basically concatenates the output of the embedding layer to provide inputs for the following clustering layer. In particular, given the set of hidden document view representations $z^1, z^2, z^3, \dots$, obtained by the embedding layer, the fusion layer is conducted by concatenating all the views representations as $[z^1, z^2, z^3, \dots]$. In the clustering layer, document clustering is adopted to learn the document partition. In our paper, we adopt the clustering approach introduced in the DEC model [25] because it provides a clustering objective that can be used to fine tune the mapping, parameterized by a deep neural network. As introduced in the DEC model, the clustering approach simultaneously optimizes both the clustering centers and the network parameters by using cluster-wise complementary semantics to improve the ESEs.

The number of clusters C is predefined. As described by the DEC model, the clustering loss is defined as the KL divergence depicted in Eq. 6:

$$L_{MDC} = KL(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (6)$$

where P is auxiliary distributions with softer probabilistic targets. Q is the data distribution of soft allocation to P . Q is obtained by Student's t -distribution. q_{ij} is the similarity between multi-view feature z_i and cluster centre μ_j . q_{ij} and p_{ij} are estimated as follows:

$$q_{ij} = \frac{(1 + \|z_i - \mu_j\|^2)^{-1}}{\sum_j (1 + \|z_i - \mu_j\|^2)^{-1}} \quad (7)$$

$$p_{ij} = \frac{q_{ij}^2 / \sum_i q_{ij}}{\sum_j (q_{ij}^2 / \sum_i q_{ij})}$$

The target distribution P is clearly defined by Q , so minimizing L_{MDC} is a form of self-training [38].

In the optimization stage, the MDCD model uses mini-batch stochastic gradient decent (SGD) and back-propagation to optimize both the clustering centre and the network parameters. The gradient of loss function L_{MDC} of composite feature z_i and μ_j of the multi-view of each data point in every view are calculated as follows:

$$\frac{\partial L}{\partial z_i} = 2 \sum_{j=1}^K \left(1 + \|z_i - \mu_j\|^2\right)^{-1} (p_{ij} - q_{ij}) (z_i - \mu_j) \quad (8)$$

$$\frac{\partial L}{\partial \mu_j} = 2 \sum_{i=1}^n \left(1 + \|z_i - \mu_j\|^2\right)^{-1} (p_{ij} - q_{ij}) (z_i - \mu_j) \quad (9)$$

Note that the MDCE model are trained with two separated stages, in particular, the pre-training and fine-tuning stage. In the pre-training stage, parameters of ESEs are learnt to discover the enhanced semantic feature representation of each document view. There may be the optimization inconsistency problem of the ESEs since they are trained separately during the pre-training stage. However, during the fine-tuning stage, the MDCD model incorporates the ESEs, the fusion layer, and clustering assignment layer in the single framework. All these three components are optimized simultaneously, which guarantees the consistence of the ESEs and MDCD results.

The time complexity of the MDCE model can also be further divided by the pre-training and fine-tuning stages. Given a reasonable set of neighbors, the pre-training results can be processed beforehand and stored for later usage. The pre-training of each NAEs and CAEs can be conducted in a parallel manner. To simplify the analysis, we assume the settings of base structure in all deep autoencoders (e.g., AE, NAE and CAE) are the same, denoted by p . m is the number of encoder layers (not including the input layer) and N is the number of samples. More precisely, the time complexity of training V m -layer NAE is of order $\mathcal{O}(VN\tau((m-1)p^2 + \omega p) + VN^2)$, where τ is the maximal number of training iterations and ω is original feature dimensions of each view x^v (assuming they are the same). Normally, $p < \omega$, so the time cost is $\mathcal{O}(VN\tau m \omega p + VN^2)$. The time complexity of training CAE is slightly more complex, which is in order of $\mathcal{O}(V^2 N \tau m \omega p + VN^2(V-1))$. In fine-tuning stage, the time cost of CAE and NAE is of order $\mathcal{O}(VN\tau m \omega p)$, cause here we only need to fine-tune the parameters of the encoder part for all views. The time complexity of cluster assignment is linear and can be calculated as $\mathcal{O}(VN\tau)$ which can be ignored.

4. Experiments

4.1. Datasets

Five real-world text corpora were used to generate 6 datasets to conduct extensive experiments. Statistics of the datasets are summarized in Table 1.

Aminer corpus¹ is a research paper dataset that is mainly used to study the research social network. Each paper is associated with its abstract, authors, year, venue, and title. We derived two text datasets from the *Aminer corpus*. The first dataset,

¹ <https://www.aminer.cn/data>.

Table 1
Statistics of the datasets.

Dataset	# of examples	# of classes	# of views
<i>Aminer</i>	4,306	3	2
<i>Aminer(700)</i>	700	3	2
<i>3-sources</i>	169	6	3
<i>Reuters</i>	600	6	5
<i>Movies</i>	617	17	2
<i>Multi-source news</i>	25,023	12	2

namely, *Aminer*, contains 4,306 randomly selected papers from three manually labelled research areas, in particular, “informatics”, “database”, and “graphics”. Each research paper is represented in two views: the content view and author view. The content view is collected from the abstract of the paper. The author view contains the authors’ names collected from the current paper and its references. All features in the two views are pre-processed by the tf-idf scheme. The second dataset, namely, *Aminer(700)*, is a subset of the *Aminer* dataset that contains 700 randomly selected papers.

3-sources² is a news articles corpus collected from 3 online news sources: BBC, Reuters, and Guardian. There are 948 news articles of 6 topic classes in the *3-sources corpus*. We chose those topics that are available in all 3 data sources from the *3-sources corpus* and derived the *3-sources* dataset for conducting experiments.

Reuters was introduced in [39]. Documents in the *Reuters corpus* are organized in 6 classes written in 5 different languages. We use each language as a document view. The *Reuters* dataset is derived by randomly sampling 100 documents from each class, resulting in a dataset of 600 documents.

Movies³ is a movie corpus extracted from IMDb. It contains 617 movies organized in 17 genres. *Movies* in the *Movies corpus* are described in two views, in particular, the keyword view and the actor view.

Multi-source news is a newly generated news dataset to conduct large scale experiments. It contains timeline news articles collected from 4 social platforms including sina.com, tencent.com, sohu.com and people.com. We derived the *Multi-source news* dataset by randomly selecting 25,023 news articles under 12 news events. Each news article is described in 2 views, in particular, the title view and content view.

4.2. Experimental setting

We compared our proposed MDCE model with 10 state-of-the-art multi-view clustering algorithms. The first 4 state-of-art models are **a series of MLRSSC models**⁴ that are developed by multi-view subspace clustering models and show good clustering performances [22], in particular, the pairwise MLRSSC model (“P-MLRSSC”), the centroid-based MLRSSC model (“C-MLRSSC”), and the kernel extensions of both the pairwise and centroid-based MLRSSC models (“P-KMLRSSC” and “C-KMLRSSC”, respectively). To investigate the performances of those models that make use of complementary information between the views, we chose 3 methods for comparison, in particular, the **FMR**⁵ [5] model, the **MSC_IAS**⁶ [23] model, and the **APMC**⁷ [36] model. **AE²-Nets**⁸ [28], **MvDSCN**⁹ [34], and **DEMVC**¹⁰ [31] were investigated for comparison since they are state-of-the-art deep clustering models for multi-view clustering. A simple combination of the normal AE model and the k-means model is regarded as the baseline denoted by **Multi-AE + k-means**.

Our proposed MDCE model does not have many parameters to tune. The structure of the deep encoder network of all ESEs is set as $\omega - 600 - 200 - 2000 - 10$. ω is the dimension of the input document view (features). The decoder part of the ESEs is a mirror of the encoder part. Except for the input, output and embedding layers, all internal layers are activated by the ReLU nonlinearity function [40]. We extend the judgment condition of clustering convergence in DEC [25]: if the difference between the clustering results of the previous two times is less than the threshold, the iteration will be stopped. We conducted a parameter tuning process for the other comparison models. The MLRSSC and its related models have 4 key parameters: μ , β_1 , β_2 , and $\lambda^{(v)}$. We chose μ from the set of values $\{10^1, 10^2, 10^3, 10^4\}$. The low-rank parameter β_1 was tuned from 0.1 to 0.9 with step 0.2, and the sparsity parameter β_2 was set to $(1 - \beta_1)$. The consensus parameter $\lambda^{(v)}$ was tuned from 0.3 to 0.9 with step 0.2. After the parameter tuning process, we chose the set of parameters with the best performance on a small set of tuning datasets and applied the set of parameters to all MLRSSC-related models. For a fairer comparison, we set the number of nearest neighbours (anchors) in MSC_IAS and APMC the same as our proposed MDCE model. K was

² <http://mlg.ucd.ie/datasets/3sources.html>.

³ <http://lig-membres.imag.fr/grimal/data.html>

⁴ <https://github.com/Geovhbn/MLRSSC>.

⁵ <https://github.com/lsrlh/FMR>.

⁶ http://www.cbsr.ia.ac.cn/users/xiaobowang/codes/MSC_IAS_Released.zip.

⁷ https://github.com/eeGuoJun/AAAI2019_APMC.

⁸ <https://github.com/willow617/AE2-Nets>.

⁹ <https://github.com/huybery/MvDSCN>

¹⁰ <https://github.com/JieXuUESTC/DEMVC>.

Table 2

Comparison of the clustering performances of the MDCE-related models, Multi-AE + k-means, P-MLRSSC, C-MLRSSC, P-KMLRSSC, C-KMLRSSC, MSC_ias, APMC, FMR, AE²-Nets, MvDSCN and DEMVC on all datasets. The performances were evaluated with NMI and CA (%).

	<i>Aminer</i>		<i>Aminer(700)</i>		<i>3-sources</i>		<i>Reuters</i>		<i>Movies</i>		<i>Multi-source news</i>	
	NMI	CA	NMI	CA	NMI	CA	NMI	CA	NMI	CA	NMI	CA
Multi-AE + k-means(baseline)	82.65	95.96	82.37	95.50	48.44	51.92	31.71	51.972	22.44	21.07	88.80	82.95
P-MLRSSC	84.97	96.70	84.26	96.00	53.60	55.74	37.51	51.82	31.35	30.28	–	–
C-MLRSSC	85.05	96.73	84.26	96.00	53.44	56.33	37.83	53.27	30.26	29.25	–	–
P-KMLRSSC	85.45	96.70	82.50	95.50	46.67	50.80	37.19	56.50	26.88	25.50	–	–
C-KMLRSSC	87.37	97.21	82.50	95.50	46.12	50.98	36.52	56.41	25.53	23.78	–	–
MSC_ias	85.47	96.80	82.37	95.50	55.13	66.86	36.50	50.22	28.28	26.42	90.20	87.99
APMC	86.26	96.96	85.60	96.50	63.44	70.41	39.32	40.00	27.95	30.47	88.59	81.80
FMR	84.63	96.24	82.42	95.67	51.11	59.59	34.90	48.57	27.51	26.71	–	–
AE ² -Nets	86.08	96.88	83.19	95.63	54.44	59.82	34.68	53.11	23.66	23.81	85.56	87.13
MvDSCN	84.54	96.59	78.33	94.57	52.25	63.91	36.09	55.33	21.52	24.80	–	–
DEMVC	87.12	97.26	83.15	95.71	49.97	44.24	32.61	51.67	22.32	23.01	91.05	89.19
MDCE-NAE	87.05	97.16	85.56	96.50	53.39	59.88	43.56	59.25	28.51	25.28	95.32	91.31
MDCE-CAE	88.37	97.44	86.63	97.00	65.04	73.43	42.06	59.08	25.09	23.66	91.65	90.77
MDCE-MIX	89.18	97.70	87.07	97.00	67.10	73.22	43.02	59.03	26.47	25.62	95.31	93.79

randomly generated within the range of values [1]. For large-scale data, it needs to be adjusted according to the data situation. The parameters of FMR are as follows: $\{\lambda_1 = 1e-3, \lambda_2 = 1e-7, \epsilon = -5e8, \alpha = 10, K = 50\}$, $\{\lambda_1 = 1e-3, \lambda_2 = 1e-7, \epsilon = -5e8, \alpha = 10, K = 300\}$, $\{\lambda_1 = 1e-3, \lambda_2 = 1e-9, \epsilon = -5e9, \alpha = 100, K = 100\}$, $\{\lambda_1 = 1e-3, \lambda_2 = 1e-9, \epsilon = -5e9, \alpha = 100, K = 50\}$, $\{\lambda_1 = 1e-3, \lambda_2 = 1e-9, \epsilon = -5e9, \alpha = 100, K = 50\}$, $\emptyset$. The order corresponds to the 6 datasets in Table 2. According to the author's suggestion, we also adjusted the appropriate parameters (such as learning rate, activation function, nodes of layers in AE) with each dataset for AE²-Nets, MvDSCN and DEMVC. Note that, the AE modules in the MvDSCN model and the DEMVC model are designed by conv2d, which is not applicable to text data. We modified them with the AE framework that is consistent with ours.

Because of the dependence of unsupervised learning on initialization, we ran all models 20 times except MSC_ias and APMC. The performance was computed by taking the average of these 20 experiments.

We evaluated clustering performance using two different metrics: normalized mutual information (NMI) and unsupervised clustering accuracy (CA). The range of all these metrics was [0,1], with a higher value indicating better performance.

4.3. Experimental results and analysis

4.3.1. Overall performance of the MDCE model

To evaluate the effectiveness of the NAE and CAE models, we investigated 3 different schemes on the embedding layer of the MDCE model: using the NAE model only ("MDCE-NAE"), using the CAE model only ("MDCE-CAE"), and using both the NAE and CAE models ("MDCE-MIX").

Table 2 depicts the experimental results for our proposed MDCE-related models and all other comparative models on all real-world datasets in detail. For the *Multi-source news* dataset, because of its large size of sample scale, the series of MLRSSC models, the FMR model, and the MvDSCN model run out of memory due to their extensive usage of computational resources. As shown in Table 2, our proposed MDCE-related models consistently outperform all other methods in terms of all tested measures. The NMI and CA performances of the MDCE-related models were at most 38% and 41% higher than those of the baseline model (the Multi-AE + k-means). The improvement on the *Aminer* dataset was smaller than those of the others because the quality of the *Aminer* dataset is better. The baseline and other models achieved relatively good results, which implies that the quality of those features in the original representation is useful for discovering document structures. It leaves little space for the complementary information to improve. However, our proposed MDCE model still performed the best on the *Aminer* dataset. The improvements on the *3-sources* and *Reuters* datasets were more substantial because the quality of the original document representation was poor in these datasets, leading to poor document clustering performance. Moreover, the MDCE-NAE performs slightly better on the *Reuters* and the *Movies* datasets. It indicates the type of complementary information contributes differently for different datasets. For the *Reuters* and the *Movies* dataset, more meaningful information are obtained from the neighbor-wise semantics other than the cross-view semantics. One reason is that nature of document views in these two datasets varies dramatically which leads to inconsistent optimal clustering assignment. Discovering additional cross-view semantic information may also introduce noise information at the same time. Depicted by those results on the *Multi-source news* dataset, deep clustering models, in particular, our proposed MDCE model and the DEMVC model, normally have more promising performances due to the ability of deep neural network on processing large-scale data. The AE²-Nets performs slightly worse because the motivation of the model is to learn a complete data representation for further usage. It is not specially designed for the clustering problem. Besides, all MDCE-related models show obvious advantages on this large-scale dataset as the neighbors of each document can be sufficiently discovered. On

Table 3

Comparison of clustering performances including MDCE-AE, MDCE-NAE, MDCE-CAE, MDCE-MIX, NAE + k-means, and CAE + k-means on all datasets. For comparison, the performance of the baseline model (Multi-AE + k-means) is also depicted. The performances were evaluated with NMI and CA (%).

	<i>Aminer</i>		<i>Aminer(700)</i>		<i>3-sources</i>		<i>Reuters</i>		<i>Movies</i>		<i>Multi-source news</i>	
	NMI	CA	NMI	CA	NMI	CA	NMI	CA	NMI	CA	NMI	CA
Multi-AE + k-means(baseline)	82.65	95.96	82.37	95.50	48.44	51.92	31.71	51.97	22.44	21.07	88.80	82.95
NAE + k-means	83.74	95.98	82.70	95.50	48.72	54.32	39.82	55.35	26.49	23.02	92.70	89.02
CAE + k-means	86.86	97.09	84.32	96.00	62.70	72.18	40.71	56.50	23.60	22.20	89.62	86.86
MDCE-AE	85.18	96.61	83.64	96.00	50.10	52.75	36.63	56.88	24.04	21.88	90.04	85.10
MDCE-NAE	87.05	97.16	85.56	96.50	53.39	59.88	43.56	59.25	28.51	25.28	95.32	91.32
MDCE-CAE	88.37	97.44	86.63	97.00	65.04	73.43	42.06	59.08	25.09	23.66	91.65	90.77
MDCE-MIX	89.18	97.70	87.07	97.00	67.10	73.22	43.02	59.03	26.47	25.62	95.31	93.79

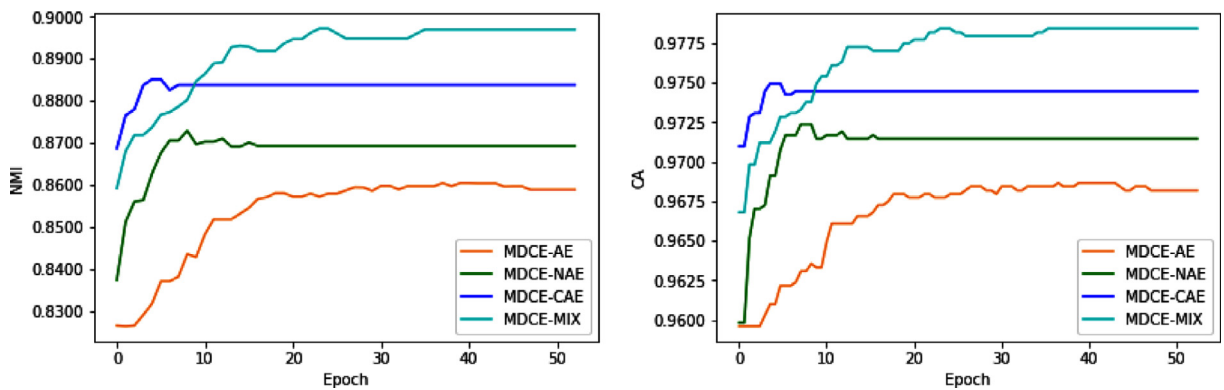


Fig. 4. Trace plot for the clustering performances, evaluated by NMI (left) and CA (right), of the MDCE-related models on the *Aminer* dataset with increasing number of training epochs.

the contrary, for those datasets with many classes but a small amount of data samples, such as *Movies*, the AE-based method is weak.

4.3.2. Performances of the variations of the MDCE Model

We investigated the performance of our proposed ESEs without the help of the clustering layer of the MCDC model. After the features were embedded and fused by the NAE and CAE models, the traditional k-means clustering model was employed to obtain the document assignment (“NAE + k-means” and “CAE + k-means”). Table 3 depicts the experimental results for variations of our proposed MDCE model and all other comparative models on six real-world datasets in detail. The performance of the MDCE model with normal AE models is also depicted for comparison.

As shown in Table 3, compared with the MDCE-AE model, our MDCE-related models (in particular, the MDCE-NAE, MDCE-CAE, and MDCE-MIX model) all performed better. Therefore, our designed NAE model and the CAE model are effective at complementing the feature semantics with neighbour-wise and view-wise semantic information. Compared with the NAE + k-means and the CAE + k-means models, the MDCE-NAE and the MDCE-CAE models all had obviously better performance. This demonstrates that the fine-tuning process of the clustering layer of the MCDC model is useful. The feature representations can be substantially improved by using the document clustering semantics. The MDCE-MIX model performed stably well on multi-view document clustering. It generally achieved the best performance compared with all other models. However, the computational time of training the MDCE-MIX model was also costly compared with the two other models. In summary, ESEs are effective at mapping and complementing the higher-dimensional document space to lower-dimensional feature representations. The clustering layer of the MCDC model is also effective at fine-tuning with cluster-wise complementary semantics.

We investigated the convergence of the MDCE-related models. The performances of all MDCE-related models with an increasing number of iterations on the *Aminer* dataset are depicted in Fig. 4. The performance of the MDCE-AE model is also depicted for comparison. The trends of the NMI and CA results are comparable. In the first few iterations, the values of all MDCE-related models increased quickly and gradually became flat. The convergence speed of the ordinary MDCE-AE was obviously slowest, with the worst clustering effect. The convergence speed of the MDCE-MIX model was slower than that of the MDCE-NAE and MDCE-CAE models because the MDCE-MIX model uses a large number of ESEs. As a result, the number

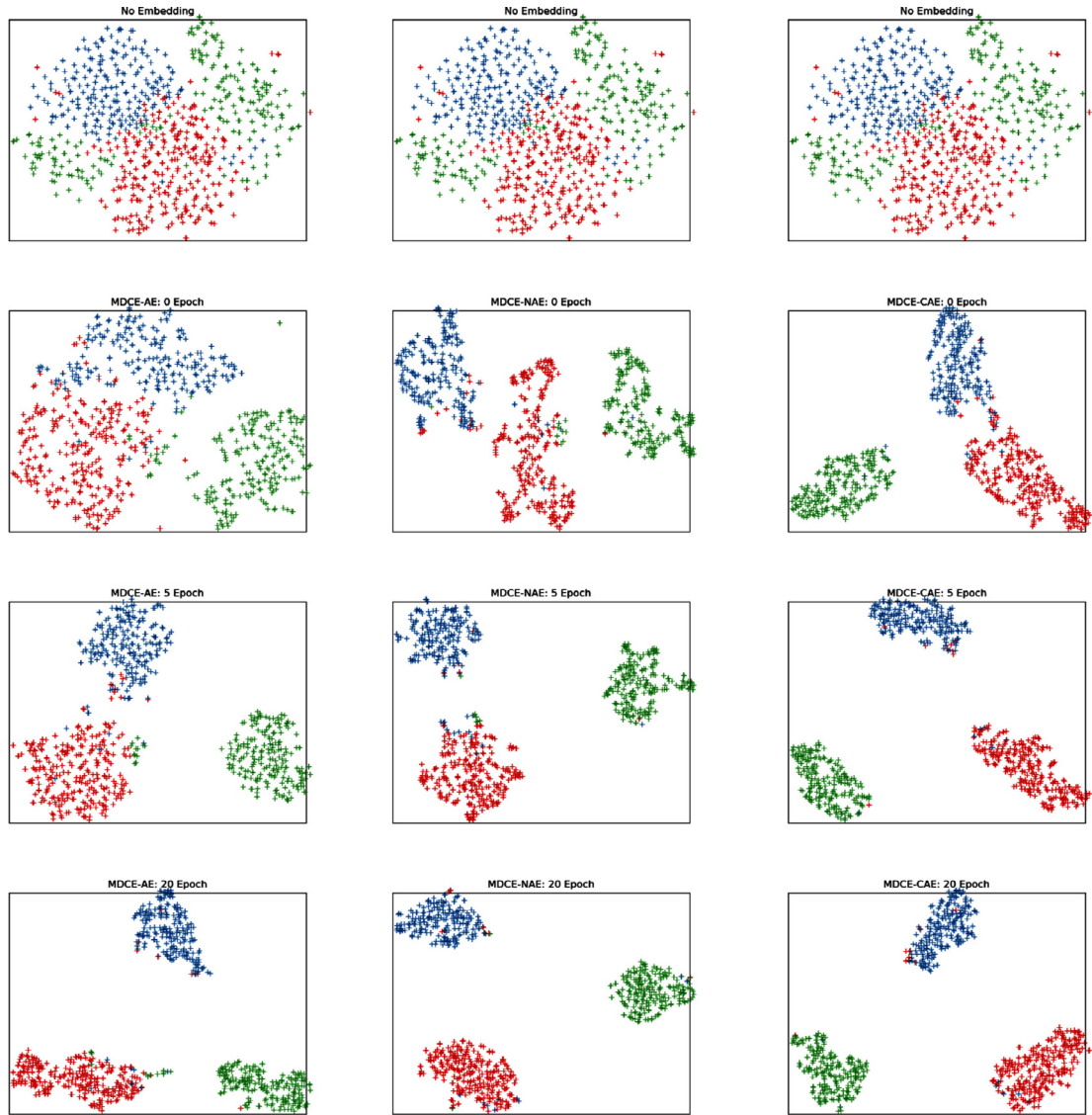


Fig. 5. Visualization of clustering structures on the *Aminer(700)* dataset. Different colours mark different clusters. Each column from left to right indicates the clustering structures of MDCE-AE, MDCE-NAE and MDCE-CAE models, respectively. The first line depicts the original structure of data samples. Each line from 2 to 4 depicts the structure of the clustering assignments achieved by different number of epochs of the clustering process.

of parameters in this model is substantially larger than that in the other models. However, with a reasonable number of epochs (the number of epochs was 50 in our experiments), the MDCE-MIX model converges to a stably good result.

4.3.3. Visualization

We visualized the document structure by t-SNE[41] on the *Aminer(700)* dataset. The visualization results are shown in Fig. 5. Each column from left to right indicates the MDCE-AE, MDCE-NAE and MDCE-CAE models, respectively. The first line in Fig. 5 depicts the documents represented in the original document space. Each line from 2 to 4 depicts the running epochs of the clustering process demonstrated in the lower-dimensional feature space obtained after the fusion layer of the MDCE model. In particular, we selected 0, 5, and 20 epochs of the clustering process for demonstration. Note that the feature space of the 0 epoch is the feature space obtained by the feature embedders with no fine-tuning of the clustering process. The clusters converge in 20 epochs. Therefore, the final results of the clustering are given in the last row of Fig. 5. As shown in Fig. 5, the clustering process clearly improves the feature representations. The documents in the original data space are loose with unclear cluster boundaries. As the number of epochs increases, the cluster property of the features becomes increasingly obvious. The cluster information is clearly displayed. All methods separate clusters well in most of the documents, but the data structure of MDCE-NAE and MDCE-CAE is better than that of MDCE-AE. In the final result of the MDCE-AE model,

Table 4

Comparison of running time (in second) of MDCE-related models, Multi-AE + k-means, P-MLRSSC, C-MLRSSC, P-KMLRSSC, C-KMLRSSC, MSC_IAS, APMC, FMR, AE²-Nets, MvDSCN and DEMVC on all datasets.

	<i>Aminer</i>	<i>Aminer(700)</i>	<i>3-sources</i>	<i>Reuters</i>	<i>Movies</i>	<i>Multi-source news</i>
Multi-AE + k-means(baseline)	219.14	43.05	28.80	293.51	22.96	90.10
P-MLRSSC	1996.15	8.60	0.47	7.58	5.36	–
C-MLRSSC	1979.58	7.91	0.47	6.80	4.99	–
P-KMLRSSC	1195.07	4.61	0.42	3.20	4.49	–
C-KMLRSSC	1183.89	4.63	0.40	2.93	4.80	–
MSC_IAS	125.59	12.65	38.79	138.11	8.30	7074.30
APMC	14.85	1.85	0.36	0.74	1.53	5232.49
FMR	2964.68	123.75	4.73	59.80	91.23	–
AE ² -Nets	265.85	34.75	37.24	434.83	19.01	2034.12
MvDSCN	265.85	34.75	37.24	434.83	19.01	–
DEMVC	265.85	34.75	37.24	434.83	19.01	296.00
MDCE-NAE	243.04	53.26	43.75	588.12	28.25	390.64
MDCE-CAE	248.56	53.83	48.04	604.06	29.20	614.10
MDCE-MIX	489.33	111.74	92.18	1102.76	55.39	892.55

Table 5

The NMI standard deviation of series of MLRSSC models and ours on *3-sources* and *Reuters*.

(a) series of MLRSSC models		
Models	<i>3-sources</i>	<i>Reuters</i>
P-LRSSC	0.0200	0.0137
C-LRSSC	0.0139	0.0125
P-KLRSSC	0.0406	0.0188
C-KLRSSC	0.0457	0.0144
(b) MDCE-related models		
Models	<i>3-sources</i>	<i>Reuters</i>
MDCE-AE	0.0394	0.0028
MDCE-NAE	0.0006	0.0111
MDCE-CAE	0.0352	0.0015
MDCE-MIX	0.0354	0.0025

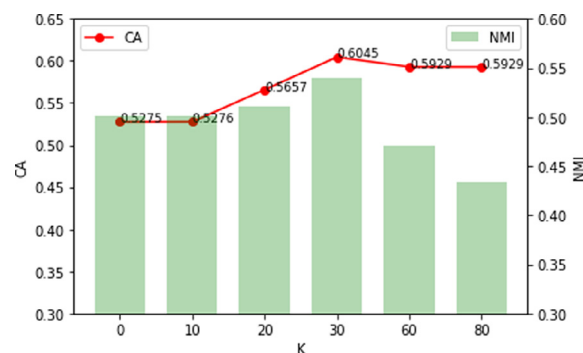


Fig. 6. The effect of the neighbour number K on the MDCE-NAE model. The broken line is the trend of CA, and the bar chart is NMI.

it is obvious that the clusters with blue and red colours are loose, and there are many green dots sticking to the edge of the red cluster. Clusters of the MDCE-NAE and the MDCE-CAE models are denser, and documents are more clearly separable.

4.3.4. Computational complexity analysis

We investigated the computational running time of the MDCE model in this part of experiments, because one concern of it is that it may need a long processing time for usage. The running time of all models are illustrated in Table 4. Experiments

of the *Multi-source news* dataset are conducted on one machine with 1 NVIDIA P40 GPU. Other experiments are conducted on another machine with ordinary computation resources¹¹. The average running time is calculated by taking the average of 20 runs of experiments.

From Table 4, the computational time needed by the deep clustering models is generally longer than other models. Our MDCE-related models need less time than the AE²-Nets model since the AE²-Nets model investigates a two-level AE-hierarchy framework to learn data representations. Compared with MDCE-related models, the DEMVC model uses less time because it doesn't need to discover the neighbor-wise and view-wise semantical information to enhance the view representations. Nevertheless, the execution time of the MDCE-related models are satisfactory in practice. All MDCE-related models can obtain reasonable results within 15 min under an appropriate setting of GPU experimental resources. The computational time should be worthwhile in exchange for significant improved performance.

4.3.5. Other analysis

There are few parameters to be initialized in our proposed MDCE model. For the initialization of the parameters in the networks, we run the MDCE-related models 20 times after pre-training the ESEs. The differences in the 20 experiments are discussed by taking the standard deviation of all experimental results evaluated by NMI measurement. The standard deviation of each model performance is shown in Table 5. The standard deviation values of our proposed MDCE-related models are obviously less than series of MLRSSC models. By comparing on the *3-sources* and *Reuters*, these models showed gaps of varying degrees. Therefore, our proposed MDCE model performed stably with network initialization.

In addition to the network parameters, another parameter that must be set initially is the number of neighbours K selected for each document to discover complementary semantics. We explored the clustering performance of our proposed MDCE-NAE model by varying the value of K . The experimental results are presented in Fig. 6. The value of K was set in the range of {0,10,20,30,60,80} on the *3-sources* dataset. As shown in Fig. 6, the document clustering performance showed rapid improvement with a small number of K . K gradually reached the optimal value. When K increased to a relatively large value, the NMI and the CA results of document clustering were flat with a slow descending tendency due to the introduction of noise information with more neighbours. However, the effect of noise was cancelled to a certain degree because the average of the neighbours was used to represent the neighbours. Therefore, our proposed MDCE model can obtain promising results with a small number of neighbours selected. K was rather stable over a wide range. Therefore, in our experiments, we randomly set the value of K for all experiments within a reasonable range.

5. Conclusion and future work

In this paper, we address the problem of multi-view document clustering. A novel model, namely, MDCE, is introduced to address the sparseness, high dimensionality, and inconsistency issues of multi-view documents. Unsupervised embedders are designed to learn and enhance mapping of high-dimensional data space to low-dimensional feature space with various types of complementary information. A deep learning network is employed to simultaneously learn the document clustering assignments and improve the mapping with cluster-wise complementary information. The experimental results demonstrate that the MDCE model is effective and performs better than other state-of-the-art multi-view clustering models.

An interesting direction for future research is to study how to use the MDCE model to handle noise features of text document views, which is also a crucial problem of document analysis. The idea of adding complementary useful semantic information can be explored to reduce those confusing noise features.

CRedit authorship contribution statement

Ruina Bai: Investigation, Writing - original draft. **Ruizhang Huang:** Conceptualization, Methodology, Writing - review & editing, Supervision. **Yanping Chen:** Validation, Resources. **Yongbin Qin:** Validation, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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¹¹ The experiments were conducted on a PC with a Intel(R) Core(TM) i7-8700 CPU and 8 GB memory.

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